SSH

1. Introduction

SSH (Secure Shell) is a secure authentication method used in Git and GitHub to connect a local machine to a remote repository. Instead of using a username and password every time.

In GitHub, SSH is commonly used to push, pull, and clone repositories securely.

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2. Why Use SSH with GitHub?

When working with GitHub, there are two main ways to connect to a repository:

- HTTPS
- SSH

Using SSH has several advantages:

- No need to enter username and password repeatedly
- More secure authentication
- Faster workflow for developers
- Better for professional environments and automation

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3. How SSH Works in GitHub

SSH uses two keys:

- **Private Key** → Stored securely on your local machine.
- **Public Key** → Uploaded to your GitHub account.

When you try to connect:

1. GitHub checks your public key.
2. Your computer proves it has the matching private key.
3. If they match, access is granted.

This process ensures secure authentication without sending passwords over the internet.

4.Commands

```
noran@Nour MINGW64 /d/ITI -Full Stack/Version Control/Day_1_Noran_Elkhalifa (master)
$ ssh-keygen -t ed25519 -C "noranelkhalifa6@gmail.com" -f ~/.ssh/id_ed25519
Generating public/private ed25519 key pair.
Enter passphrase for "/c/Users/noran/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/noran/.ssh/id_ed25519
Your public key has been saved in /c/Users/noran/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:Y8Rl2dhxTxnCu2gAUhUbb9ZiD36p4nSg7VcgeCOOpnY noranelkhalifa6@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|      ...++=oo..+|
|     . o o* +o.+ |
|      . =. B .. .|
|       + ==.+..  |
|      o S.+.o+.  |
|     o oo..ooo   |
|    o  . +.o.    |
|   o E  + o.     |
|  . .    o.      |
+----[SHA256]-----+

noran@Nour MINGW64 /d/ITI -Full Stack/Version Control/Day_1_Noran_Elkhalifa (master)
$ ls -la ~/.ssh
total 10
drwxr-xr-x 1 noran 197609   0 Mar  3 16:38 ./
drwxr-xr-x 1 noran 197609   0 Mar  3 16:35 ../
drwxr-xr-x 1 noran 197609   0 Mar  3 16:35 agent/
-rw-r--r-- 1 noran 197609 419 Mar  3 16:38 id_ed25519
-rw-r--r-- 1 noran 197609 107 Mar  3 16:38 id_ed25519.pub

noran@Nour MINGW64 /d/ITI -Full Stack/Version Control/Day_1_Noran_Elkhalifa (master)
$ ssh-add ~/.ssh/id_ed25519
Identity added: /c/Users/noran/.ssh/id_ed25519 (noranelkhalifa6@gmail.com)

noran@Nour MINGW64 /d/ITI -Full Stack/Version Control/Day_1_Noran_Elkhalifa (master)
$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIFqs3h3Ok2MMiB9TKLw+NkeWDCzup7Sy8HVO5kL7oIdb noranelkhalifa6@gmail.com

noran@Nour MINGW64 /d/ITI -Full Stack/Version Control/Day_1_Noran_Elkhalifa (master)
$ ssh-add -l
256 SHA256:Y8Rl2dhxTxnCu2gAUhUbb9ZiD36p4nSg7VcgeCOOpnY noranelkhalifa6@gmail.com (ED25519)

noran@Nour MINGW64 /d/ITI -Full Stack/Version Control/Day_1_Noran_Elkhalifa (master)
$ ssh -T git@github.com
The authenticity of host 'github.com (140.82.121.3)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi NoranKh132! You've successfully authenticated, but GitHub does not provide shell access.

noran@Nour MINGW64 /d/ITI -Full Stack/Version Control/Day_1_Noran_Elkhalifa (master)
$ |
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